

Bristol Myers Squibb Associate Software Engineer - What to Learn

Why these keywords

This role sits at the intersection of Decision Intelligence (a platform called Aera) and AI/ML/Gen AI engineering. Your RAG, vector-embeddings, and agentic-AI project work (Echo, Vibe) is a strong, genuine match for the AI/ML half of the JD. The delta is almost entirely the Aera platform itself (proprietary, unlikely to be taught anywhere outside BMS) and two named agentic-AI frameworks (LangGraph, Agent SDK) plus more formal ETL/ data-engineering vocabulary.

MUST learn (Class A – explicitly required)

Aera (Decision Intelligence platform) – a proprietary enterprise platform for building “decision flows” that turn data into automated business decisions/actions. No public sandbox exists – it’s learned on the job. 20%: it’s like a low-code workflow/automation builder, where a “Skill” is a packaged use case and a “crawler” is a scheduled data-ingestion job. Be ready to say “no hands-on Aera experience, but I’ve built ETL-style pipelines (Echo’s document ingestion into per-tenant vector namespaces is functionally an ETL pipeline).”

LangGraph – a framework for building stateful, multi-step agentic workflows as a graph of nodes (LLM/tool calls) connected by edges (control flow/branching). Bridge: a formalized version of the agentic loop you already hand-built in Vibe (Inngest AgentKit). 20%: build one simple LangGraph app with 2-3 nodes and a conditional edge.

Agent SDK (likely OpenAI’s Agents SDK) – a toolkit for defining agents with tools/handoffs/guardrails, similar in spirit to the Zod-typed tool-calling pattern in Echo. 20%: skim the quickstart – concepts (agent, tool, handoff) map closely to what you already built.

ETL & data modeling/validation – formal Extract-Transform-Load vocabulary and data-quality practices. You’ve done this informally (document ingestion for RAG, Mixpanel event sync) without the formal terms. 20%: know the E/T/L stages by name and what “reconciliation” means (source vs. destination consistency checks).

GOOD to know / already strong (Class A/B, genuine strength)

RAG, vector embeddings, semantic search – you’re already strong here (Echo: Gemini embeddings, 3072-dim, Convex vector search). Lead with this in interviews – it’s the JD’s core AI/ML ask.

LLM APIs, prompt engineering, agentic AI workflows – also a genuine strength (Echo’s 3-tool support agent, Vibe’s 15-iteration agentic loop). Frame your existing work explicitly as “agentic AI” using the JD’s own language.

AWS, Docker, Git – already used (AWS Secrets Manager in Echo, Docker-built E2B sandbox in Vibe, Git daily). No new prep needed – just be ready to name the AWS Secrets Manager specifics if asked.

Priority if time is short

1. LangGraph basics (most concrete, most learnable in a short time, directly named in the JD).
2. Aera – just enough conceptual framing to talk about it intelligently (can’t get hands-on before an interview, so don’t over-invest).
3. ETL/data-reconciliation vocabulary (quick vocabulary pass, low effort).
4. Agent SDK – lower priority, conceptually closest to what you know.